What's New In LED And LUMINAIRE TEST EQUIPMENT

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LED and LED equipment manufacturers are upgrading test equipment and related software to get substantial data related to principal parameters of LEDs such as intensity, luminous flux, colour, spatial radiation and pattern. This will help them with more reliable and accurate results.

For instance, in LED manufacturing process, variation in the brightness or colour between two LEDs of the same batch is common. Thus, LEDs need to be tested to ensure the quality at component and end-product level.

Test equipment for LED luminaires

Electrical and photometric tests of luminaires are based on standards defined by authorised bodies such as The International Electrotechnical Commission (IEC), Underwriters Laboratory (UL), Bureau of Energy Efficiency (BEE) and European Commission (EU).

Photometric test equipment measure the lumen, intensity, temperature and colour coordinates of a luminaire, while electrical test equipment measure the input and output current, voltage and power parameters. These tests ensure ruggedness and reliability of the hardware at the manufacturing end and safety and cost-effectiveness of the product at the customer end.

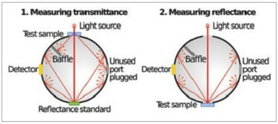

Fig. 1: Integrating sphere concept (Image courtesy: www.photometricsphere.com)

LED test equipment and enhancements

A renowned LED luminaire manufacturer had been using the integrating sphere frequently for lumen tests. But soon after a year, it started facing issues with the reflective coating of integrating sphere. The feedback was given to the test equipment manufacturer, which decided to recoat the sphere. It was a time-taking procedure, so the customer had to wait for four months to get the sphere back in shape.

Such issues are faced by most manufacturers. To help them provide cost-effective yet reliable solutions, test equipment manufacturers have released several updates in their instruments as well as technology.

Increasing reflectivity ratio in integrating sphere

Jason Zhu, regional manager-business development, Everfine Corporation, says, "India is an emerging market for LED business, so equipment manufacturers focus primarily in this zone."

Zhu says that manufacturers have released integrating spheres with reflectivity ratio increased up to 98 per cent because properties of the coating material have been upgraded. Ever since the material has changed, there are hardly any customer complaints about the reflective coating peeling off. Thus, the upgrade in material has not only benefitted

Fig. 2: Integrating sphere (Image courtesy: www.ul.com)

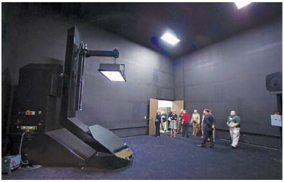

Fig. 3: Mirror-type goniophotometer (Image courtesy: www.ul.com)